

Training Configuration and Summary:

Sequencing:

- timesteps: 30
- batch_size: 128

Model Hyperparameters:

- min_units: 128
- max_units: 256
- units_step: 64
- dropout_min: 0.2
- dropout_max: 0.3
- dropout_step: 0.1
- activation_functions: ['tanh', 'relu', 'elu']
- optimizers: ['adam', 'rmsprop', 'Nadam']

Tuning:

- max_epochs: 30
- hyperband_iterations: 3

LSTM Multi-Step Training Report

Report Generated: 2025-04-14 11:48:04

Training Start: 2025-04-14_08-23-03

Training End: 2025-04-14_11-36-14

Training Duration: 193.17 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: tanh

Optimizer: Nadam

Dropout Rate: 0.20

Performance Metrics (Multi-Step Prediction):

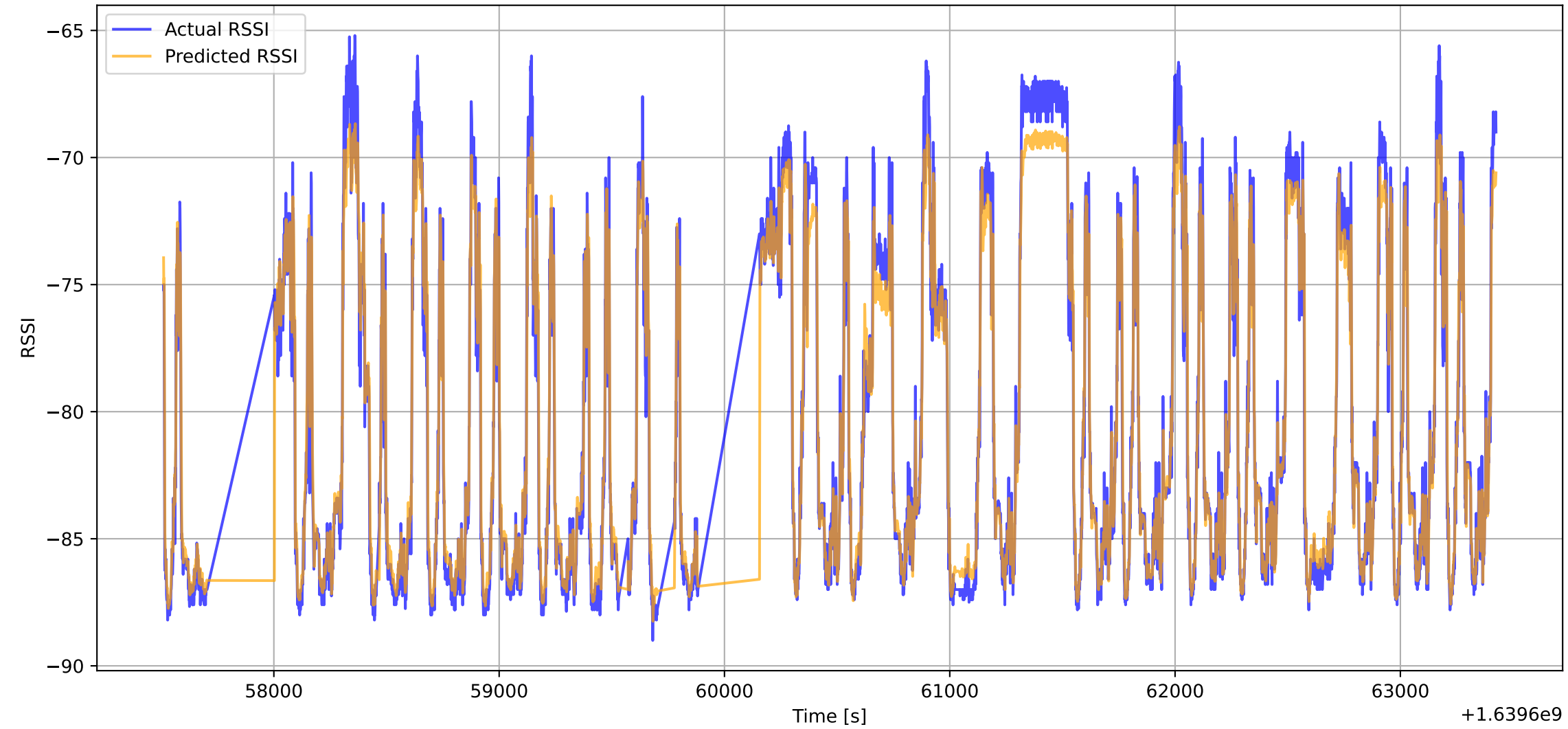
MAE: 1.1183

MSE: 2.6197

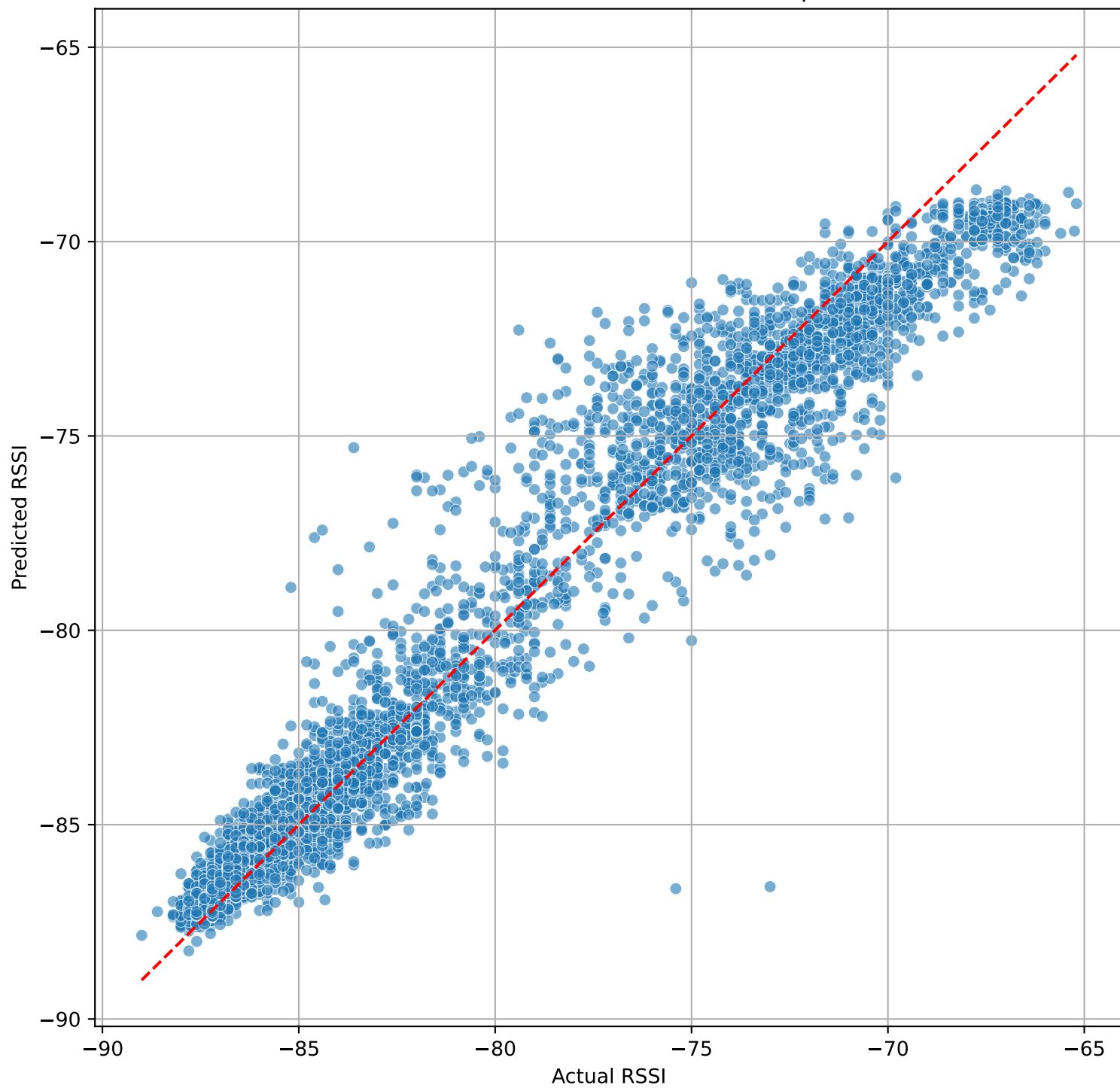
RMSE: 1.6185

R2 Score: 0.9406

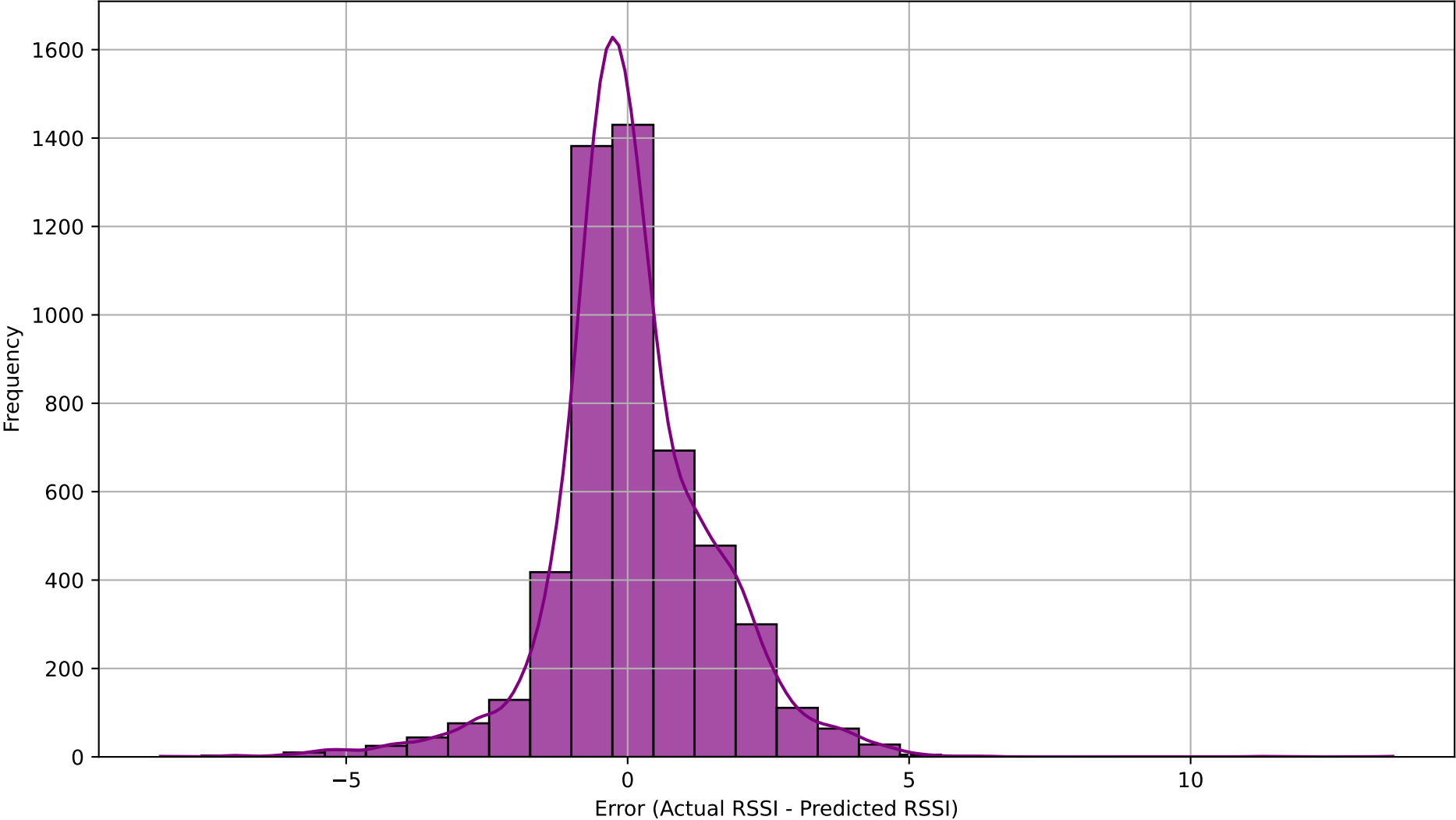
Actual vs Predicted RSSI (First Step)



Actual vs Predicted RSSI (First Step)



Distribution of Prediction Errors (First Step)



Predicted RSSI in Spatial Context (First Step)

